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Journal:	*BMJ Open*
Manuscript ID	bmjopen-2019-032915
Article Type:	Research
Date Submitted by the Author:	11-Jul-2019
Complete List of Authors:	Xiao, Yunyu; New York University, Silver School of Social Work Pinkney , Edward; University of Hong Kong, Centre for Suicide Research and Prevention Au, Terry Kit Fong; University of Hong Kong, Department of Psychology; University of Hong Kong, President's Office Yip, Paul; University of Hong Kong, Department of Social Work and Social Administration; University of Hong Kong, Centre for Suicide Research and Prevention
Keywords:	gender diversity, gender equity, higher education, Athena SWAN

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Athena SWAN and Gender Diversity: A UK-based Retrospective Cohort Study

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Word count: 3,221

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Abstract

Objectives: To promote gender diversity and equity in higher education, the Athena SWAN (Scientific Women's Academic Network) supports and recognizes higher education institutions (HEIs) in advancing the careers of women through awards, a charter of commitment, and associated training and advocacy since 2005. Most of the existing evaluation studies, however, focus on qualitative assessments. This study sought to examine the relationship between Athena SWAN participating HEIs in the UK and gender diversity of leaders and senior academics using quantitative data over five years (2012/13-2016/17). It also explored the associations between Athena SWAN awards and university performance.

Design: Retrospective cohort study based on the UK HEIs.

Setting: 148 HEIs provided by the Higher Education Statistics Agency (HESA).

Participants: Female-to-male ratios of the employees were collected for each academic year from 2012/2013 to 2016/2017.

Primary and secondary outcome measures: Data concerning female representation rates of positions in leadership (e.g., heads of institutions, department heads) and professor (i.e., full professors without line management responsibilities) were evaluated separately. QS Ranking overall scores were collected to measure university performance. Analysis of variance with Bonferroni corrections and *t*-test were used to examine differences in female representation across Athena SWAN status.

Results:

Gender diversity of leaders and professors in all levels of Athena SWAN status has improved over the five years. Athena SWAN awardees had lower female representation than non-awardees in senior academic staffs but showed accelerating progress towards gender equity in leadership positions ($p<.05$). Silver-award institutions ranked higher in QS rankings than Bronze-award institutions ($t=1.79$, $p<.10$).

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Conclusions: Athena SWAN has limited influences on gender diversity during the five years of examination, especially among senior-level academic positions. Future research is encouraged to examine the effect of Athena SWAN on gender equity across more extended periods. More actions are warranted to promote institutional desire to foster gender diversity.

Keywords: gender diversity, equity, higher education, Athena SWAN

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Strengths and Limitations of this Study

- This study prospectively investigates the impact of an internationally adopted gender equity programme (Athena SWAN) in 148 high education institutions (HEIs) in the UK.
- This is one of the first studies that examined gender diversity in leadership and academic positions across HEIs with different Athena SWAN status in a longitudinal design.
- We showed that gender diversity of leaders and professors increased over the five years (2012/13-2016/17), while the Athena SWAN levels had limited effects.
- The effect of Athena SWAN manifested in the gender diversity of leaders in HEIs but not professors.
- Among Leaders in HEIs, Athena SWAN level was positively associated with progress based on both the rate of change and narrowing gaps between groups. Notably, Silver Awardees pushed past the point of one-third of leaders being female; a figure previously suggested to be an essential milestone for how leadership board diversity affects organizational performance.
- The study was limited by not including controlled variables, and only having access to five years of data. Future research could add further insight through controlling other variables and looking over a longer period.

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Introduction

Gender inequality has been persistent in academic medicine and collegial governance.[1-6] Despite increasing policies that aims at reducing gender imbalance, under-representation of women in the higher education institutions (HEIs) continues to exist, particularly among the senior levels of administrative leaders and professors.[3 7 8] In the United Kingdom (UK), only 21% of clinical academics were women and such gender inequality increases with seniority, with 8.4% of all staff being female professors in 2017.[8] Similarly, among key leadership roles across 173 HEIs in the UK, 27% of governing bodies are chaired by a woman, and only 29% have a woman Vice-Chancellor or Principal.[5]

Under-representation of women in HEIs is associated with individual and institutional costs across different levels. First, women were more likely than male faculty members to perceive discrimination in promotion, resource, salary, and fellowship, creating obstacles to satisfaction and retention of the women in HEIs.[9] Such experience could be a threat to their physical and mental health,[10] and could even bring intergenerational effects on child health.[11] Second, losing qualified women to participate in leading clinical practice and academic research, while having more than half female medical students entering the medical schools,[12] is considered a waste of public investment in human capital.[13] Finally, the shortage of female researchers participating in the research agenda of clinical medicine may bring implicit bias in the research agenda of clinical medicine and public health, which could further impede the international competitiveness of a country's translational research.[7 14]

HEIs have a broad set of responsibilities to society, as educators, as research bodies, as employers, and as civic institutions.[15] The impact of HEIs can be measured

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using narrower metrics such as graduate achievements and research influence, as well as broader socially-oriented measures that indicate a commitment to social values such as societal wellbeing, diversity and equal opportunity. These broader measures of social value can be challenging to quantify and are typically not recognized in rankings of institutions.[16] As a result, there is a role for independent frameworks and marks of recognition to support and incentivize institutions to prioritize social responsibilities. To promote gender diversity and equity, as well as addressing the adverse effects of gender inequity in academic sciences, the Athena SWAN (Scientific Women's Academic Network) Charter was established, aiming to address the unequal representation of women and to encourage commitment to advancing the career opportunities for women in both higher education and research.[17]

Managed by the Equality Challenge Unit (ECU) since its launch in 2005, Athena SWAN offers HEIs who join the Charter the opportunity to demonstrate excellence in supporting gender diversity practices by progressing through Bronze, Silver and Gold Awards. Both Bronze and Silver Awards involve collection and submission of data, while Bronze Awards recognize detailed plans for making improvements and Silver Awards recognize retrospective achievements and milestones. As of March 2019, no educational institutions had attained Gold Awards.

In 2011, the National Institute for Health Research (NIHR) announced that from 2016 only institutions in receipt of an Athena SWAN Silver Award would be shortlisted for research funding awards. In 2012, these funding awards were worth upwards of £100,000 to some institutions, creating a substantial incentive for institutions with medical schools to pursue awards.[18] Initially focused on careers of women in Science, Technology, Engineering, Mathematics and Medicine (STEMM), the scope of Athena SWAN has expanded in 2015 to cover other disciplines, non-

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academic staff, trans-sexual staff, and students. The influence and success of the initiative in signing up the majority of HEIs in the UK has inspired similar initiatives internationally, such as in Australia.[19]

Several studies have attempted to evaluate the influences of Athena SWAN, while results have been inconsistent. Some studies identified positive effects.[3 4 6] For instance, a report commissioned by ECU found that 90% of institutional champions (facilitators appointed within institutions) and 81% of departmental champions agreed that Athena SWAN had a positive impact on gender equity.[20] Over half of Athena SWAN Champions demonstrated promoting progress on female career opportunities. Academic staffs in HEIs with Silver Award were more satisfied with their "career performance/development review with opportunities for training and development" than staffs in departments without the Award.

Munir and colleagues also found evidence of improvements in institutional engagement and communication around diversity practices after receiving an Athena Award.[20] Members of participating institutions reported "visible representation of more women in key positions and senior roles." Concerning reasons for applying for an Athena SWAN Award, the majority of HEIs reported they thought it was the "right thing to do" and that they were committed to gender equality.

Other studies addressed the adverse effects associated with the initiative. Using qualitative data among institutions implementing the Athena SWAN, Caffrey and colleagues concluded that while gender equity programs have the potential to address inequity, paradoxically, they can also reinforce inequities unless more macro-level institutional practices supported the programs, national policies, and societal norms to enhance gender equity.[3] This is partially due to the reinforcement of women's family responsibility while gender equity programmes emphasizing on family-friendly

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policies. Since women have undertaken a disproportionately higher amount of work in Athena SWAN, the process of achieving the award was less sustainable and systematic.[2]

Furthermore, a recent exploratory study interviewing of staffs in university medical science departments concluded that while Athena SWAN was perceived to support women's career development and contribute to challenging discrimination, broader cultural changes within institutions and society as a whole were necessary to address fundamental gender equality issues.[1] Besides, the connection between the Award and NIHR funding could pervert incentives and lead to overemphasis on getting an Award at the expense of fundamental and systematic advancements in gender diversity and support.

Most of the existing studies use qualitative design. The only quantitative study of Athena SWAN used employment data from university medical schools between 2004 and 2013.[18] The study found that institutions affected most by the NIHR announcement appeared to be more motivated to attain Athena SWAN Awards, but there was no significant difference in the rates of female employment (including lecturers, senior lecturers, and professors) over time between those institutions that were signed up to the Athena SWAN Charter and those that were not.[18] However, the study was limited by the fact that study period ended three years before the NIHR deadline that required funding applicants to have a Silver Award, which meant the effects of any institutions that subsequently pushed to meet the deadline were excluded. To date, the relationship between Athena SWAN Awards and female employment has remained unclear.

This study attempts to address the gaps in the existing knowledge and to explore the relationship between Athena SWAN affiliation levels and staff diversity in senior

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roles. Diversity in senior higher education ranks and academic positions, as an indicator of how well women are progressing through career stages, is a key measure within the higher education equity and diversity agenda, and hence, an essential indicator of the success of Athena SWAN. Importantly, the study period includes the five years up to and including the NIHR funding requirement in 2016. Besides, to understand the university performance among different levels of Athena SWAN recipients, this study examines whether Athena SWAN Award levels are associated with the QS World University Rankings.

Methods

Data and Measures

HEI employment data were provided by the Higher Education Statistics Agency (HESA), categorized by gender, for each academic year from 2012/2013 to 2016/2017. Female-to-male ratios were rounded to the nearest five persons to comply with data protection laws. Female representation rates of leadership and professor groups were calculated separately. '*Leaders'* refers to heads of institutions, and heads of academic areas, centers and senior functions, including professors that head departments (HESA contract levels A-E2). '*Professors'* refers to full professors without line management responsibilities (HESA contract level F1).

HESA contract levels were as follows,[21]

A: Head of Institution - Vice-Chancellor/Principal/equivalent

B: Deputy Vice-Chancellor/Pro Vice-Chancellor/Chief Operating Officer/Registrar/Secretary

C1: Head/Director of major academic area

C2: Director of major function/group of functions e.g. finance, corporate services, HR

D (1-3): Head of a distinct area of academic responsibility center e.g. head of

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school/division/department

E1: Head of a subset of an academic area/director of a small center (may include professors who head departments/research center)

E2: Senior function head

F1: Professor (without line management responsibilities)

Examining both leaders and professors allowed for exploring academic versus leadership/managerial career pathways independently. Since many higher education leaders have progressed from their roles as professors, this study could also consider different stages of careers in HEIs.

Athena SWAN Award recipients and Charter members are published and maintained on the ECU website and were collected in February 2019. While Awards are given on both an institutional level and departmental level, only institutional level Awards were used for this study. No institutions had Gold level Awards; therefore for this study, there were four levels of Athena SWAN accreditation: Bronze, Silver, non-recipient Charter institutions (i.e., had committed to the Charter but had not received an Award), and 'Non-Athena' institutions without affiliation to Athena SWAN.

This study included 148 HEIs. By the time of data collection, the sample consisted of 111 Athena SWAN members, among which 77 (52.0%) were Bronze institutions, 18 (12.2%) were Silver institutions, and 16 (10.8%) were non-recipient Charter institutions. The rest of HEIs (*n*=37) were Non-Athena institutions. QS Overall scores in 2017 (indicators for QS ranking) were used as a proxy for university performance. While imperfect, QS rankings reflect the global reputation of institutions.[22]

Statistical Analysis

Descriptive statistics were used to show the trends in female representation in

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leaders and professors groups among all HEIs and across Athena SWAN levels from 2012/13 to 2016/17 (Table 1, Figure 1, and Figure 2). Secondly, *t*-test was conducted to compare the differences in the percentages of female in leadership and professors in 2016/17 across Athena and non-Athena charter members (Table 2), followed by analysis of variance (ANOVA) with Bonferroni corrections for multiple comparisons to examine the differences in female representation (%) of 2016/17 leaders and professors among the various Athena SWAN levels (Table 2). To examine the influence of Athena SWAN awards on university performance, *t*-test was used to compare the overall scores of QS rankings in 2016/17 between Silver and Bronze awardees (Table 3). Listwise deletion were used to deal with missing data.

Results

Descriptive Statistics

Table 1 presents the percentages of women leaders and professors in HEIs from 2012/13-2016/17. Female representation in both groups has increased over this period. On average, there were higher percentages of female leaders (*Mean* = 39.5%, *Standard Deviation [SD]* = 0.09) than professors (*Mean* = 36.3%, *SD* = 0.10) across the five years. Between 2012/13 and 2016/17, the percentages of female leaders increased by 6.6% (from 62.9% to 69.5%), or 10.5% in relative terms. The percentage of female professors increased by 5.1% (from 34.0% to 39.1%), or 15.0% in relative to the 2012/13 baseline.

[INSERT TABLE 1 HERE]

Changes of Female Representation over Time

As shown in Figure 1, over the five years, non-recipient Charter institutions had higher percentages of female leaders than Non-Athena institutions. Surprisingly, non-recipient Charter institutions also had higher percentages of female leaders than both

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Bronze and Silver institutions.

[INSERT FIGURE 1 HERE]

Over the five academic years examined, gender diversity in leadership positions has improved across all HEIs, while different rates of changes were observed for different Athena SWAN levels (see Table 1). Silver institutions were moving towards gender equity at a faster rate (16.7%) than Bronze institutions (7.4%), non-recipient Charter institutions (5.3%), and non-Athena institutions (0.1%). The gaps in gender diversity between the groups have also narrowed proportionately, suggesting that Athena SWAN levels may be associated with better overall progress in moving towards gender equity in leaders.

As seen in Figure 2, non-Athena institutions had higher percentages of female professors than Athena institutions regardless of whether their Athena levels. For professors, Silver awardees had the highest rate of increasing gender diversity (13.9%) over institutions from other levels. Bronze Awardees had a lower rate of change (10.4%) than non-recipient Charter institutions. Non-Athena institutions had the lowest rate of change (2.1%, see Table 1).

Notably, the gap in gender diversity between Silver and non-recipient Charter institutions grew over the period (4.7% in 2013, 5.3% in 2017). Bronze Awardees had a slightly higher gender diversity than non-recipient Charter institutions in 2012/2013 (0.4%), while the pattern reversed in 2016/17 (difference = -0.4% in 2017), suggesting that the progress of approaching gender diversity among Bronze Awardees had fallen slightly behind non-recipient Charter institutions by 2016/2017. In sum, a higher Athena SWAN accreditation level did not seem to be associated with better overall progress in moving towards gender parity for professors.

[INSERT FIGURE 2 HERE]

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Gender Diversity among Leaders

Using the latest sample employment data (2016/2017) results in Table 2 indicated there were non-significant differences in Athena and non-Athena members in female representation in leadership positions (t = 1.23, p = .22). However, there was a significant difference in female representations in leaders across Athena SWAN levels (F=5.15, df=3, p =.002). Among Athena SWAN members, Bronze (t=-2.84, p<.05) and Silver institutions (t=-3.65, p<.01) had significantly lower percentages of female leaders than those in non-recipient Charter institutions. Silver institutions also had a marginally significant lower female representation than the non-Athena institutions (t=2.66, p<.10).

[INSERT TABLE 2 HERE]

Gender Diversity among Professors

Using the latest same employment data (2016/2017) results in Table 2 revealed non-significant differences in female representation in professor positions between Athena and non-Athena institutions, as well as across Athena SWAN levels (F=2.13, df=3, p=.10).

Athena Award Level and University Performance

Results concerning the differences in the performance rankings of institutions with Awards were marginally significant (p<.10, See Table 3), where Silver institutions ranked higher in QS rankings than Bronze institutions (t = 1.79). There was only one non-recipient Charter institution covered in QS rankings, suggesting that most of the institutions considered to be globally elite have applied for (and received) Athena SWAN Awards, with the highest-ranked institutions mostly having attained Silver Awards.

[INSERT TABLE 3 HERE]

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Discussion

A central goal of Athena SWAN is to move HEIs and the sector towards gender equity. With this in mind, the increasing trends in gender diversity found in this study are primarily encouraging. In all levels of Athena SWAN affiliation, and even in those without affiliation, gender diversity of leaders and professors increased over the five years (2012/13-2016/17). Among leaders in HEIs, Athena SWAN level was positively associated with progress based on both the rate of change and narrowing gaps between groups. Notably, Silver Awardees pushed past the point of one-third of leaders being female; a figure previously suggested to be an essential milestone for how leadership board diversity affects organizational performance.[23]

For professors (i.e., full professors without managerial duties), the findings are mixed. While all four groups moved towards greater gender diversity, in the most recent year of data (2016/2017), less than one-third of professors were women. Over the five years, institutions with Bronze Awards slipped behind non-recipient Charter institutions, and the gap in diversity widened with institutions with Silver award lagging further behind non-recipient Charter institutions. Thus, Athena SWAN levels generally seem to be a poor indicator of progress towards gender diversity amongst academics.

We also found that having an Athena SWAN Award is associated with a higher global ranking of UK HEIs. As suggested in previous research,[18] this can be partly explained by institutions that are most affected by the NIHR funding being motivated to pursue Silver Awards. Higher-ranking institutions may also have better financial resources to commit to the Athena SWAN initiative. Such factors have contributed to debate about whether Awards are recognizing what they ought to achieve. The latest single-time diversity figures (2016/2017) showed that the highest Athena SWAN

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level was associated with lower gender diversity in both the leaders and professors groups. Coupled with the professor diversity trends amongst Athena SWAN Awardees, a question emerged to whether Awards are recognizing the right institutions.

The relatively few females in the professor group and slow rate of improvement in gender diversity are particularly striking because around half of postgraduate research students are female.[24] The slow rate of increase can be partly explained by there being more positions to fill in academia than in leadership roles, meaning it takes more time to achieve equity. Nevertheless, other potential factors, such as the gender makeup of students graduating in different subject areas (e.g., liberal arts vs. STEM subjects), may also affect the speed of progress towards gender diversity in the professor group. Future studies may consider how gender differences in subject areas may influence progress towards diversity.

Strengths and Limitations

This study is one of the first studies to examine the gender diversity in leadership and academic positions across UK HEIs with different Athena SWAN status in a longitudinal design. It contributes to previous literature, among which most have a qualitative focus,[1 3] through quantitatively assessing the relationship between these variables.

The following limitations should be considered when interpreting our results. First, while we were able to obtain five-year longitudinal data from 2012/13-2016/17, the effect of Athena SWAN may differ in longer periods. It is possible that changes in female representation, especially among the senior levels, require longer periods to appear. Hence, future studies that continue to investigate this issue across more years are warranted. Second, the status of Athena SWAN was based on the most recent

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records in 2019. However, it is possible that different HEIs were awarded different levels of Athena SWAN (or withdrawals of awards) across the study periods. It would be meaningful to record the exact date for the Athena SWAN award status for each HEI in order to examine the short-term and long-term effect of Athena SWAN on gender diversity. Third, gender diversity in HEIs is affected by various social contexts (e.g., culture, policy, geography).[1-6 9] As the first study to explore this issue, we did not control these variables in the current analysis. Future research would be encouraged to consider these factors while analyzing the effect of Athena SWAN. Lastly, this study is based on HEIs with available employment data from HESA in the UK, the generalizability of the study results across the national and to other countries have not been determined. Future studies is encouraged to explore how other gender equity programs, such as Organizational Change for Gender Equity in STEM Academic Professions (ADVANCE) in the United States, has effects on gender diversity in other countries.

Conclusion

Under-representation of women in senior positions among the higher education institutions has been widely identified as a considerable problem to address; both in the UK and internationally. This study sheds light on the relationship between Athena SWAN level and gender diversity among higher education institutions. Despite overall positive trends towards diversity, and positive associations between Athena SWAN level and improvements in leadership diversity over time, the findings amongst professors indicate that Athena SWAN may still need to consider whether receiving the Award could bring a meaningful and sustained commitment to gender diversity. The relationship between research funding and criteria to achieve Athena SWAN levels may also warrant further attention to ensure appropriate institutional

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recognition and incentives. Finally, to address gender diversity in a broader cultural context and the degree to which the climate of the higher education environment is supportive of diversity, Athena SWAN might incorporate social marketing initiatives aimed at attitudinal change, such as the Time to Change campaign[25] and its work in tackling mental health stigma.

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Table 1

Trends of Female Representation in Leaders and Professors from 2012/13 to 2016/17[a.]

Year	Leaders (%)						Professors (%)					
	N	All HEIs	Bronze	Silver	Non-recipient Charter institutions	Non-Athena	*N*	All HEIs	Bronze	Silver	Non-recipient Charter institutions	Non-Athena
2012/13	136	62.9	36.0	29.1	43.1	41.2	100	34	24.8	19.7	24.4	31.4
2013/14	136	62.9	35.6	29.7	43.5	41.5	106	34.7	24.9	19.9	24.3	32.2
2014/15	137	65.4	37.4	30.9	43.4	40.5	103	35.6	25.3	21.0	26.6	36.7
2015/16	135	67	37.8	33.4	45.5	40.1	102	38.2	26.3	21.9	27.7	33.9
2016/17	133	69.5	38.6	34.0	45.4	41.2	103	39.1	27.3	22.4	27.7	32.1
Rates of Changes [b.]		10.5	7.2	16.8	5.3	0.0		15.0	10.1	13.7	13.5	2.2

Note:

[a.] Rates based on the availability of data in each year.

[b.] Rates of changes were calculated using the percentage in 2017 minus the percentage in 2013 and divided by the percentage in 2013.

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Table 2

Comparisons of Female Representations in Leaders and Professors in 2016/2017 among Athena SWAN Silver, Bronze, Non-recipient Charter Institutions and Non-Athena Institutions (n = 133 for leaders and n = 103 for professors)

Categories of SWAN Award	% Female		Contrast (*t* statistic)			
	M	*SD*	Bronze	Silver	Non-recipient Charter Institutions	Non-Athena
			Leaders[a.]			
Athena						1.23
Bronze (*n* = 73)	38.7	9.1	–	–	–	–
Silver (*n* = 14)	34.0	4.4	-1.89	–	–	–
Non-recipient (*n* = 15)	45.4	9.3	-2.84*	-3.65**	–	–
Non-Athena (*n* = 31)	41.2	7.7	-1.43	-2.66†	1.58	–
			Professors[b.]			
Athena						1.36
Bronze (*n* = 70)	27.3	8.9	–	–	–	–
Silver (*n* = 17)	22.4	4.1	-2.07	–	–	–
Non-recipient (*n* = 10)	27.7	10.1	-0.12	-1.51	–	–
Non-Athena (*n* = 5)	32.1	15.8	-1.16	-2.16	-0.91	–

Note: † $p<.10$, *$p<.05$, **$p<.01$, ***$p<.001$.

[a.] *t-test* for comparing Athena vs. Non-Athena institutions is non-significant ($p = .22$). *ANOVA* comparing all levels of institutions is significant, with $F=5.15$, $df=3$, $p=.002$).

[b.] *t-test* for comparing Athena vs. Non-Athena institutions is non-significant ($p = .18$). *ANOVA* comparing all levels of institutions is non-significant, with $F=2.13$, $df=3$, $p = .10$).

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Table 3

Comparisons of QS Overall Scores in 2017 between Athena SWAN Silver and Bronze Institutions (n=46)

Institution Performance	Silver Awardees			Bronze Awardees			*t*
	n	M	SD	*n*	M	SD	
QS Overall Score 2017	16	64.0	21.8	30	53.2	18.2	1.79†

Note: t-test for comparisons between Athena SWAN Silver and Bronze recipients.
† $p<.10$, *$p<.05$, **$p<.01$, ***$p<.001$.

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Contribution

Conception and design of study: YX, EP, TKFA, PSFY. Data collection: EP. Data analysis: YX. Interpretation of data: YX, EP, PSFY. Drafting the manuscript: YX, EP. Revising the manuscript critically for important intellectual content: YX, EP, TKFA, PSFY. Approval of the version of the manuscript to be published: YX, EP, TKFA, PSFY.

Competing interests: None declared.

Funding: This research received funding from the University of Hong Kong.

Data sharing statement: Data are available upon reasonable request made to the corresponding author.

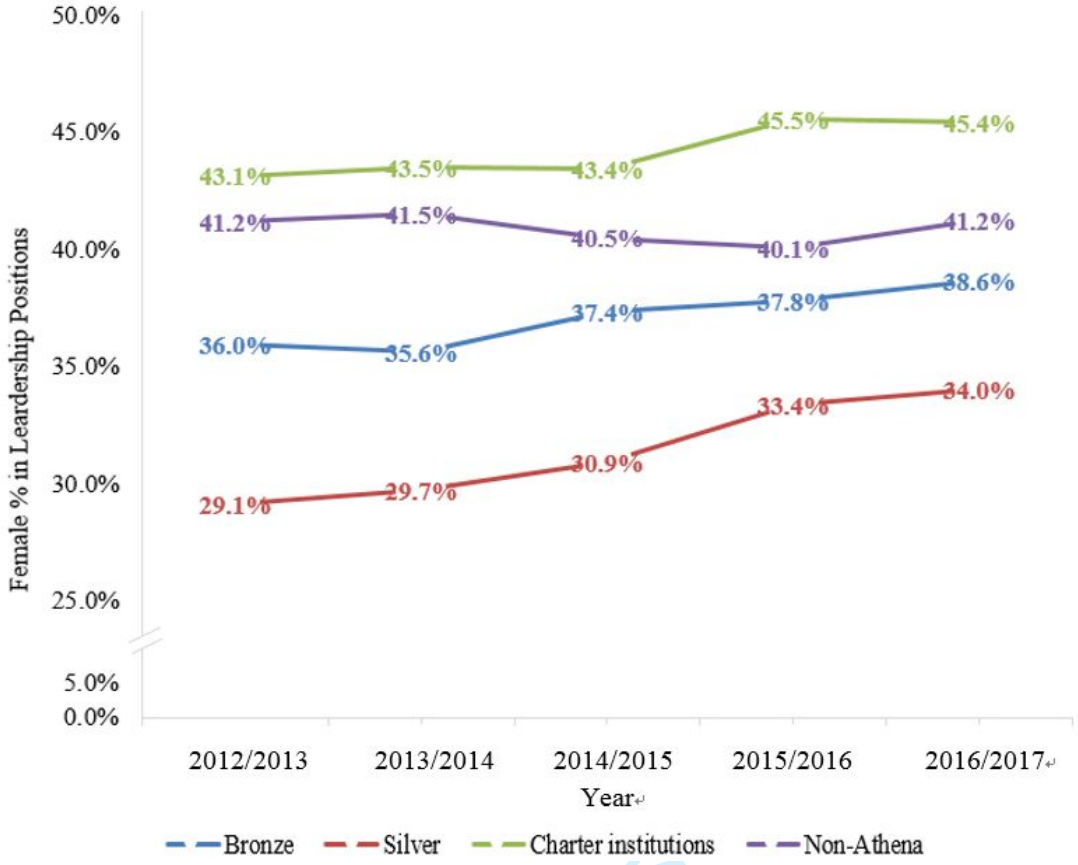
Patient consent for publication: Not required.

Provenance and peer review: Not commissioned; externally peer reviewed.

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Figure 1

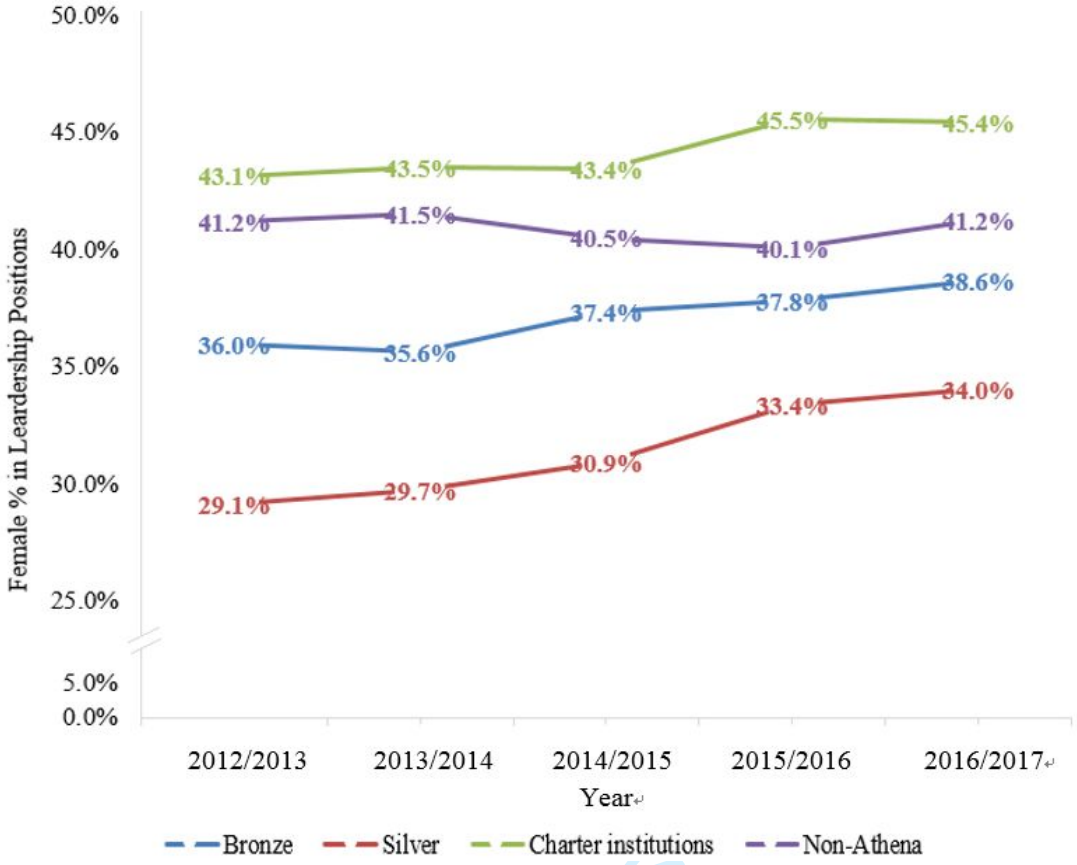
Changes of Female Representation in Leaders by Athena SWAN levels from 2012/13 to 2016/17



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Figure 1

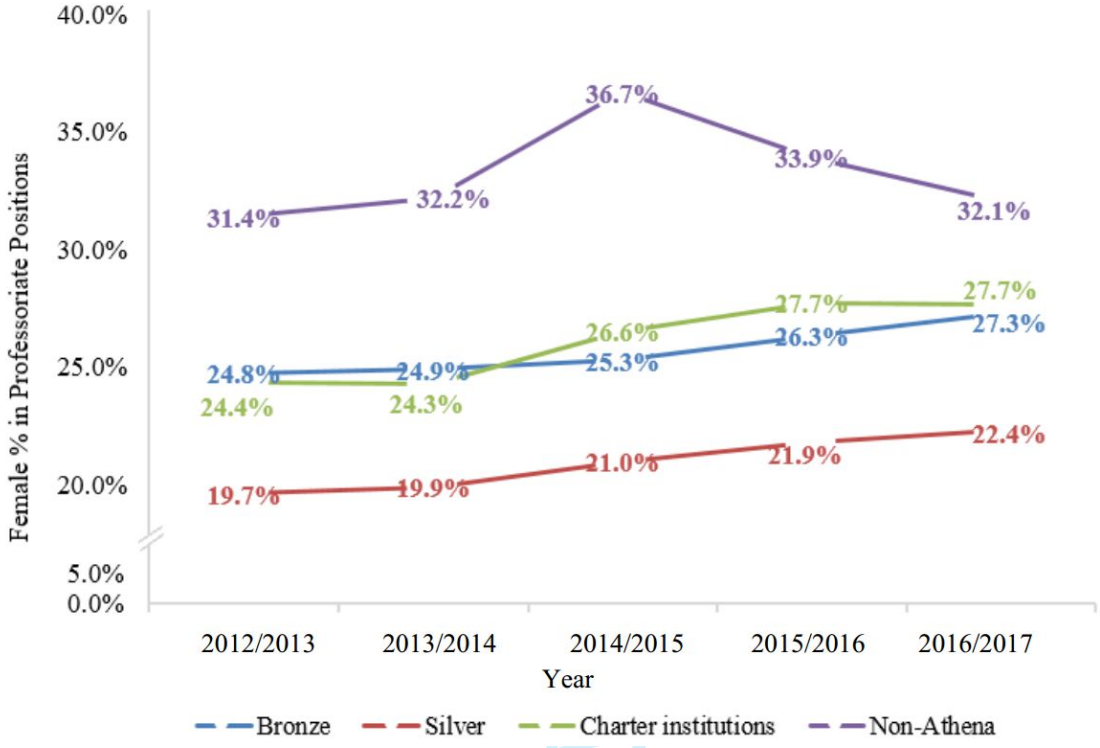
Changes of Female Representation in Leaders by Athena SWAN levels from 2012/13 to 2016/17



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Figure 2

Changes of Female Representation in Professors by Athena level from 2012/13 to 2016/17



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STROBE 2007 (v4) Statement—Checklist of items that should be included in reports of *cohort studies*

Section/Topic	Item #	Recommendation	Reported on page #
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	1-2
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	2-3
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	5-8
Objectives	3	State specific objectives, including any prespecified hypotheses	8-9
Methods			
Study design	4	Present key elements of study design early in the paper	2-3, 9
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	9-10
Participants	6	(*a*) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up	9-10
		(*b*) For matched studies, give matching criteria and number of exposed and unexposed	9-10
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	9-10
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	9-10
Bias	9	Describe any efforts to address potential sources of bias	9-10
Study size	10	Explain how the study size was arrived at	9-10
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	9-10
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	10-11
		(*b*) Describe any methods used to examine subgroups and interactions	10-11
		(*c*) Explain how missing data were addressed	10-11
		(*d*) If applicable, explain how loss to follow-up was addressed	/
		(*e*) Describe any sensitivity analyses	/
Results			

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Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	9-11
		(b) Give reasons for non-participation at each stage	/
		(c) Consider use of a flow diagram	/
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	11
		(b) Indicate number of participants with missing data for each variable of interest	20
		(c) Summarise follow-up time (eg, average and total amount)	
Outcome data	15*	Report numbers of outcome events or summary measures over time	11
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	11-13
		(*b*) Report category boundaries when continuous variables were categorized	/
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	/
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	11-13
Discussion			
Key results	18	Summarise key results with reference to study objectives	14
Limitations			
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	15-16
Generalisability	21	Discuss the generalisability (external validity) of the study results	16
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	24

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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BMJ Open

Athena SWAN and Gender Diversity: A UK-based Retrospective Cohort Study

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2019-032915.R1
Article Type:	Original research
Date Submitted by the Author:	11-Dec-2019
Complete List of Authors:	Xiao, Yunyu; New York University, Silver School of Social Work Pinkney , Edward; University of Hong Kong, Centre for Suicide Research and Prevention Au, Terry Kit Fong; University of Hong Kong, Department of Psychology; University of Hong Kong, President's Office Yip, Paul; University of Hong Kong, Department of Social Work and Social Administration; University of Hong Kong, Centre for Suicide Research and Prevention
Primary Subject Heading:	Medical education and training
Secondary Subject Heading:	Health policy, Public health, Sociology, Medical education and training
Keywords:	gender diversity, gender equity, higher education, Athena SWAN, MEDICAL EDUCATION & TRAINING

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Athena SWAN and Gender Diversity: A UK-based Retrospective Cohort Study

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Abstract

Objectives: To promote gender diversity and equity in higher education, Athena SWAN supports and recognizes higher education institutions (HEIs) in advancing the careers of women through charter commitment, awards, training and advocacy since 2005. Most evaluation studies, however, are based on qualitative assessments. This study sought to (1) examine the relationship between Athena SWAN accreditation/awards in the UK and gender diversity of leaders and senior academics using quantitative data from 2012/13 to 2016/17 and (2) explore the associations between Athena SWAN awards and university performance as measured by overall scores in the global ranking systems.

Design: Retrospective cohort study based on the UK HEIs.

Setting: Higher education sector in the UK provided by the Higher Education Statistics Agency.

Participants: 148 HEIs who provided employment data on female-to-male ratios (55% complete data) for each academic year between 2012/13-2016/17.

Primary and secondary outcome measures: Gender diversity, defined as female representation rates of positions in managerial leadership (e.g., heads of institutions, department heads) and professors. The Quacquarelli Symonds (QS) World University Ranking scores, an indicator of research, teaching, employability, and internationalization, were collected to measure university performance.

Results:

Gender diversity of managerial leaders and non-managerial professors at all levels of Athena SWAN status has improved over the five years. Linear mixed effect models identified that Athena SWAN awardees had lower female representation than non-awardees in managerial leadership positions (p <.05), while the gap was narrowed among Silver awardees over time. Athena SWAN Charter members had increasingly higher female representation than those

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not in the Charter (p <.05). Silver-award institutions ranked higher in QS rankings than Bronze-award institutions (β=11.80, p <.05).

Conclusions: There are overall rising trends in gender diversity from 2012/13 to 2016/17. Athena SWAN members showed greater and faster growth in female representations. Silver awardees had greater university performance than Bronze awardees.

Keywords: gender diversity, equity, higher education, university ranking, Athena SWAN

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Strengths and Limitations of this Study

- This is the first quantitative study to retrospectively examine gender diversity in leadership and academic positions across higher education institutions with different Athena SWAN status in a longitudinal design using five-year data collected from the Higher Education Statistics Agency (HESA).
- We showed that gender diversity of managerial leaders and non-managerial professors across all HEIs increased over the five years (2012/13-2016/17).
- Levels of Athena SWAN accreditations/awards were positively associated with the increasing rate of change in female representation, while the extents of relationship differ across groups.
- Silver awardees had better university performance measured by QS world ranking overall scores than Bronze awardees over time.
- Limitation includes lack of controlled variables and data longer than five years.

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Introduction

Gender inequality has been persistent in academic medicine.[1-6] Athena SWAN (Scientific Women's Academic Network) gender equality scheme, which was launched in 2005 by the Equality Challenge Unit (ECU) in the United Kingdom (UK) and currently managed by Advance HE, aims to support a more supportive, inclusive and flexible working environment, with less discrimination.[7] Athena SWAN has become an influential global movement, with expansion to Ireland and Australia since 2015,[8] and, recent adaptations of similar models in the United States (US) and Canada. Despite increasing policies designed to reduce gender imbalance, under-representation of women in higher education institutions (HEIs) continues to exist, particularly among the senior levels of managerial leaders and professors.[3 9 10] In the UK, only 21% of clinical academics were women, and such gender inequality increases with seniority, with only 8.4% of all staff being female professors in 2017.[9] Similarly, among key managerial leadership roles across 173 HEIs in the UK, 27% of governing bodies are chaired by a woman, and only 29% have a woman Vice-Chancellor or Principal.[5] This study focuses on HEIs in the UK due to it being the original country of Athena SWAN and providing a useful sample size of HEIs for evaluating the impact of Athena SWAN.

Gender equality has an important impact on higher education and research. Under-representation of women in HEIs is associated with individual and institutional costs across different levels. First, among faculty members, women were found to be more likely than men to perceive discrimination in promotion, resource, salary, fellowship, and scientific awards, creating obstacles to satisfaction and retention of the women in HEIs.[11] Experiences of gender bias and discrimination could be a threat to women's physical and mental health,[12] and may even bring intergenerational

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effects on child health.[13] Second, women are underrepresented as research participants in university communities. Lack of female research participants and ignorance of gender differences can result in inaccurate research conclusions.[14] Since women may respond differently to diseases, gender bias could ultimately affect the effectiveness of medical treatment choices.[15] Third, losing qualified women in the pipeline for leaders in clinical practice and academic research, while having more than half female medical students entering the medical schools,[16] is considered a waste of public investment in human capital.[17] Finally, the shortage of female researchers participating in the research agenda of clinical medicine may bring implicit bias to the fields of clinical medicine and public health.[9 18] Accordingly, researchers have recently suggested including gender balance in assessing research impact of HEIs.[14]

HEIs have a broad set of responsibilities to society, as educators, as research bodies, as employers, and as civic institutions.[19] The impact of HEIs can be measured using academic metrics (e.g., graduate achievements, research influence), as well as broader socially-oriented measures that indicate a commitment to social values (e.g., societal wellbeing, diversity, and equal opportunity). These broader measures of social value can be challenging to quantify and are typically not recognized in rankings of institutions.[20] As a result, there is a need for independent frameworks and marks of recognition to support and incentivize institutions to prioritize social responsibilities.

To address unequal representation of women and its adverse effects in academic sciences, as well as advancing the career opportunities for women in higher education and research,[21] the Athena SWAN Charter was established in 2005. Managed by the ECU since its launch, Athena SWAN offers HEIs who join the Charter the

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opportunity to demonstrate excellence in supporting gender diversity practices by progressing through Bronze, Silver, and Gold awards. Both Bronze and Silver awards involve collection and submission of data, while Bronze awards recognize detailed plans for making improvements and Silver awards recognize retrospective achievements and milestones. As of March 2019, no educational institutions had attained Gold awards.

In 2011, the National Institute for Health Research (NIHR) announced that from 2016 onwards, only institutions in receipt of an Athena SWAN Silver award would be shortlisted for research funding awards. In 2012, these funding awards were worth upwards of £100,000 to some institutions, creating a substantial incentive for institutions with medical schools to pursue Athena SWAN awards.[22] Initially focused on careers of women in Science, Technology, Engineering, Mathematics and Medicine (STEMM), the scope of Athena SWAN expanded in 2015 to cover other disciplines, non-academic staff, trans-sexual staff, and students. Recently, researchers demonstrated that Athena SWAN can be usefully framed as complex social interventions embedded in a complex system.[23] Acknowledging, planning, and evaluating Athena SWAN's contextually-embedded systematic approach is imperative to initiate effective gender equality structural and cultural changes.

Several studies have attempted to evaluate the influences of Athena SWAN, and results have been mixed. Some studies identified positive effects.[3 4 6] For instance, a report commissioned by ECU found that 90% of institutional champions (facilitators appointed within institutions) and 81% of departmental champions agreed that Athena SWAN had a positive impact on gender equity.[24] A recent mixed-method study indicated that wide-spread development and implementation of Athena SWAN plans linked to the NIHR funding incentives in medical science could benefit the perception

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of cultural inclusiveness among women in the HEIs.[25] Over half of Athena SWAN Champions demonstrated progress in promoting female career opportunities. Academic staff in HEIs with Silver award were more satisfied with their "career performance/development review with opportunities for training and development" than staffs in departments without the award. Munir and colleagues also found evidence of improvements in institutional engagement and communication around diversity practices after receiving an Athena award.[24] Members of participating institutions reported "visible representation of more women in key positions and senior roles." On reasons for applying for an Athena SWAN award, the majority of HEIs reported they thought it was the "right thing to do" and that they were committed to gender equality.

Other studies revealed unintended and sometimes adverse consequences, which perhaps highlight the complexity and challenges of implementing the Athena SWAN recommendations. First, the tendency for women to be appointed by institutions as Athena SWAN leads, and the large time commitment and labor-intensive activities associated with meeting award criteria, may, paradoxically, hamper the research trajectory of those women involved.[2] Concerns have been raised about adding administrative burden to institutions during the application process.[26] Second, the connection between the award and NIHR funding could pervert incentives and lead to overemphasis on meeting specific award criteria (i.e., box-ticking) at the expense of fundamental and systematic advancements in gender diversity and support.[2] Third, overlooking the patterns of intersectionality (i.e., the intersection of gender with ethnicity) may result in insufficient consideration of the unique disadvantages experienced by certain subgroups.[26] In fact, recent studies suggested that while gender equity programs have the potential to support women's career development and

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address gender inequity, they can also reinforce particular inequities.[3 27] It is crucial to address fundamental gender equality issues, enhance family-friendly policies, and reinforce cultural changes through high-level institutional support, along with national policies and addressing of societal norms.[28] The partially inconsistent evaluation results call for further studies to evaluate the effectiveness of Athena SWAN awards.

Existing studies almost exclusively used a qualitative design. The only quantitative study of Athena SWAN used employment data from university medical schools between 2004 and 2013.[22] That study found that institutions affected most by the NIHR announcement appeared to be more motivated to attain Athena SWAN awards, but there were no significant differences in the rates of female employment (including lecturers, senior lecturers, and professors) over time between those institutions that signed up for the Athena SWAN Charter and those that did not.[22] A key limitation of the study was that its data sampling period ended three years before the 2016 NIHR deadline that required funding applicants to have a Silver award. In other words, the effects of any institutions that subsequently met the deadline were excluded. To date, the relationship between Athena SWAN awards and female employment has also remained unclear.

The present study attempted to address the major gaps in existing knowledge and to explore the relationship between Athena SWAN affiliation levels and gender diversity in managerial leadership roles and senior academics. Specifically, this study examined the effects of levels of Athena SWAN award and the changes of female representation in 1) managerial leadership positions, and 2) professors focusing on academic duties. Moreover, to understand the effects of Athena SWAN recipients on the university performance, this study examined 3) whether Athena SWAN award levels are associated with the QS World University Rankings.

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Diversity in senior managerial and academic positions, as an indicator of the progress of women achieve through their career paths, is a key measure within the higher education equity and diversity agenda, and hence an essential indicator of the success of Athena SWAN. Importantly, the five-year study period included the rollout of the NIHR funding requirement in 2016. Based on the criteria for receiving Athena SWAN award and a recent evaluation report,[29] we hypothesized that Athena SWAN awardees would show increasing changes of female representation in managerial leaders and non-managerial professors. In addition, we hypothesized that recipients of a Silver award would have better overall scores of QS ranking than those receiving a Bronze award.

Methods

Data and Measures

Gender Diversity

HEI employment data were provided by the Higher Education Statistics Agency (HESA), categorized by gender, for each academic year from 2012/13 to 2016/17. Female-to-male ratios were rounded to the nearest five persons to comply with data protection laws. Female representation rates of managerial leadership and professor groups were calculated separately. This study focuses on HEIs in the UK since Athena SWAN was originally implemented in the UK, and we were able to collect enough employment data from HEIs to conduct longitudinal evaluation across five years. For ease of exposition and avoiding over, '*Leaders'* refers to managerial leaders including heads of institutions, academic areas, centers, and key administrative functions; this category included professors with major managerial leadership such as heading departments (HESA contract levels A-E2). '*Professors'* refers to full professors without these line management responsibilities (HESA

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contract level F1).

HESA contract levels were as follows,[30]

A: Head of Institution - Vice-Chancellor/Principal/equivalent

B: Deputy Vice-Chancellor/Pro Vice-Chancellor/Chief Operating Officer/Registrar/Secretary

C1: Head/Director of major academic area

C2: Director of major function/group of functions, e.g., finance, corporate services, HR

D (1-3): Head of a distinct area of academic responsibility center, e.g., head of school/division/department

E1: Head of a subset of an academic area/director of a small center (may include professors who head departments/research center)

E2: Senior function head

F1: Professor (without line management responsibilities)

Examining both managerial leaders (A to E) and professors without line management responsibilities (F1) allowed for exploring managerial leadership versus academic career pathways independently. Since many higher education leaders have progressed from their roles as professors, any contrast in findings between the groups may also reflect different stages of careers in HEIs.

Athena SWAN Awards

Information about Athena SWAN award recipients, Charter members, and action plans are published and maintained on the ECU and institutional websites.[31] To ensure accuracy, two authors (YX, EP) individually collected and validated the award history during the study period (2012/13-2016/17). Athena SWAN is based on 10 key principles and contains three levels of awards (Bronze, Silver, and Gold).[32] Bronze

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institution awards requires an assessment of gender equality (quantitative and qualitative), a 4-year action plan based on the assessment, and an organizational structure (e.g., self-assessment team) to implement the proposed actions. Silver awards recognize institutions that have successfully implemented the previously proposed actions since receiving a Bronze award and demonstrated the impact of Athena SWAN activities. Both the institutions and departments within the institutions can apply, while departmental awards require evidence of identifying challenges and planning actions in addition to the institutional levels.[1 32] Since this study focused on gender diversity in HEIs; only institutional-level awards were examined. No institutions had Gold level awards to date, hence for this study, there were three levels (Non-recipients Charter institutions who had committed to the Charter but had not received an award ='0', Bronze ='1', Silver ='2') of Athena SWAN accreditation for Athena Charter Members and four levels (Non-Athena='0', Non-recipients='1', Bronze='2', Silver='3') for all HEIs.

This study included 148 HEIs. By the last year of the study period (2016/17), the sample consisted of 111 Athena SWAN members, among which 79 (53.4%) were Bronze institutions, 14 (9.5%) were Silver institutions, and 16 (12.2%) were non-recipient Charter institutions. The rest of HEIs (*n*=37) were Non-Athena institutions.

Duration of Award

Since institutional efforts to achieving gender equity might show effects on female representation cumulatively since their participation in the Athena SWAN program, this study also took the duration of award (ranges from 0 to 5), measured as the length of year from the first award till the last year of study period (2016/17), into account.

University Performance

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The QS World University Rankings overall scores (ranging from 0 to 100) were employed to assess the university performance. Compared to the other world ranking systems, among which the Academic Ranking of World Universities list (ARWU) predominantly focuses on faculty accomplishments and research output,[33] the ranking of the Centre for Science and Technology Studies (CWTS) is mainly based on bibliometric data,[34] and the Times Higher Education World University Rankings (THE) focuses an institution's performance across 13 indicators, including teaching, research, knowledge transfer and international outlook,[35] the QS World University Rankings measures six simple metrics across academic reputation, employer reputation, faculty/student ratio, citations per faculty, international faculty ratio, and international student ratio.[36] While imperfect, QS rankings reflect the global reputation of institutions,[37] provided a more holistic assessment of university performance rather than focusing on bibliometric indicators, and have been shown to be consistent and valid across years. Hence, the QS Overall scores (indicators for QS ranking) were used to measure university performance.

Statistical Analysis

First, descriptive statistics were used to report the trends in female representation in Leaders and Professors groups among all HEIs and across Athena SWAN levels of accreditation/award from 2012/13 to 2016/17 (Table 1, Figure 1, and Figure 2). *Rate growth* was calculated by (1) calculating the relative female representation among Leaders/Professors of specific Athena SWAN award in specific year (i); (2) determining the relative increase or decrease in the female representation among Leaders/Professors of specific Athena SWAN award between two years (this can be computed by expressing how much [as a percentage] the female representation has increased or decreased compared to its value in the year before [i-1]); and (3) adding

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the relative increased or decreased values to determine the overall growth of the underlying rates of change during our five-year data collection period. The following equation was used:

$$Rate\ growth = \sum_{i=1}^{n-1} \frac{r_{i+1} - r_i}{r_i}$$

where $r_i = \frac{n_i}{N_i}$, n_i is the percentage of female representation for the i^{th}year and N_i is the highest possible frequency value for the specific Athena AWAN award in that year (i.e., total number of institutions received the specific Athena SWAN award). The same method was applied to describe the trend of change rate for QS overall score among all HEIs and across Athena SWAN award levels (i.e., Bronze and Sliver) of accreditation/award from 2012/13 to 2016/17 (Table 1).

Second, *t*-test was used to compare the overall scores of QS rankings in each study year between Silver and Bronze awardees (Table 1). *Cohen's d* was used to measure effect size.

Third, linear mixed effect models were employed to examine the effect of Athena SWAN accreditation/award levels (i.e., non-Athena members, non-recipients, Bronze, Silver awards) on female representations in Leaders and Professors among Athena charter members and all HEIs after adjusting for time effect ('year'). As institutions may differ in their duration of receiving and developing action plans for gender equality, duration of award was adjusted statistically (Table 2). To examine whether the effects of Athena SWAN award levels on female representation would be different with time increased, two-way interaction terms between Athena SWAN award levels and year was added to the mixed effect models. Independent variable 'year' was centralized before the creation of interaction term to reduce multicollinearity (Table 3). The same method was applied to examine the effect of

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Athena SWAN award levels (Bronze or Silver) on QS overall scores (Table 4). To examine possible nonlinear effects of time, we conducted additional analyses treating 'year' as discrete variables and using the baseline (2012/13) as the reference year. Results were shown in the Appendix. Goodness-of-fit for each model was estimated based on Akaike Information Criteria (AIC) and Bayesian Information Criteria (BIC; Singer and Willett. 2003). Smaller values of each of these statistics indicate better model fit. Statistical manipulations were undertaken using STATA software version 14.0 (Stata Corp, College Station, Texas, USA) and SAS software version 9.4 (SAS Institute, Inc., Cary, NC).

Patient and public involvement

No patients or members of the public were involved in this study.

Results

Descriptive Statistics

Table 1 presents the distributions of women managerial leaders and non-managerial professors in HEIs, as well as QS overall scores from 2012/13-2016/17. On average, there were higher percentages of female leaders than professors across the five years. Female representation in leaders was consistently the highest among non-recipients and the lowest among Silver awardees. For female professors, non-Athena institutions had the highest percentage, whereas Silver awardees showed the lowest.

The average QS overall scores increased across years from 2012/13 (mean=57.75 [SD=20.88]) to 2015/16 (mean=63.15 [SD=18.73]), but decreased in 2016/17 (mean=56.93 [SD=19.95]). Institutions with Silver awards had significantly higher QS overall scores than those with Bronze awards (p <.05) in each year across the study period.

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[INSERT TABLE 1 HERE]

Changes of Female Representation over Time

As shown in Figure 1, over the five years, non-recipient Charter institutions had higher percentages of female leaders than Non-Athena institutions. Surprisingly, non-recipient Charter institutions also had higher percentages of female managerial leaders than both Bronze and Silver institutions.

[INSERT FIGURE 1 HERE]

Over the five academic years examined, gender diversity in leadership positions has improved across all HEIs. As shown in Table 1, the rates of female increased 8.4% among managerial leaders and 7.6% among professors without line management duties, while different growth rates were observed for different Athena SWAN levels of accreditation/award. Silver institutions were moving towards gender equity at a faster rate (20.3%) than non-recipient Charter institutions (11.1%), Bronze institutions (9.0%), and non-Athena institutions (2.4%). The gaps in gender diversity between the groups have also narrowed proportionately, suggesting that Athena SWAN levels of accreditation/award may be associated with better overall progress in moving towards gender equity in managerial leadership.

As seen in Figure 2, non-Athena institutions had higher percentages of female non-managerial professors than Athena institutions, regardless of their Athena SWAN levels of accreditation/award of award. For professors, non-recipient Charter institutions had the highest rate of increasing gender diversity (17.5%) over institutions from other levels, which was followed by the growth rate of Silver awardees (14.3%) and Bronze awardees (5.7%). Non-Athena institutions had the lowest rate of change (1.1%, see Table 1).

[INSERT FIGURE 2 HERE]

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Athena Award Level and Gender Diversity

Table 2 reports the results of linear mixed effect model of gender diversity in managerial leaders and non-managerial professors. There were positive effects of year on female representation in both managerial leaders and non-managerial professors across all Athena SWAN accreditation/awards (Table 2). With time increased, the female representation increased in both managerial leaders and non-managerial professors, even without considering the effect of Athena SWAN. Among Athena SWAN charter members, differences across Athena SWAN award levels were observed only in managerial leadership positions, but not among non-managerial professors. Being Bronze (β=-.062, p <.05) and Silver (β =-.112, p <.01) awardees were associated with decreases of female leaders, compared with non-recipient Charter institutions.

Among all institutions, Athena SWAN Charter members, though not receiving awards during the study period, had significantly greater percentages of female leaders (β=.050, p <.05) than non-Athena institutions. Athena SWAN levels of accreditation/award was not associated with differences in female representation among non-managerial professors. When considering the nonlinear effects, results showed that compared to the 2012/13 baseline, percentages of women in both managerial leaders and non-managerial professors has increased since 2014/15 (Appendix 1).

[INSERT TABLE 2 HERE]

Adding the interactions between year and the Athena SWAN levels of accreditation/award revealed significant interaction effects on female representation in professor positions (Table 3). Among all HEIs, while being a Silver awardee had lower percentage of female managerial leaders than non-Athena institutions in the

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beginning, that gap was narrowed with the increase of years (β=.0014, p <.05). Results of the nonlinear effect models revealed that the buffering effects of time started at 2015/16 (See Appendix 2). While Silver awardees showed lower rates of female professors than non-Athena institutions (β=-.114, p <.05), these patterns have been slower significantly from 2015/16 to 2016/17 (β=.058, p<.01).

[INSERT TABLE 3 HERE]

Athena award Level and University Performance

Table 4 indicated that Silver awardees, in relative to Bronze award recipients, had greater QS overall scores (β=11.8, p <.05). In addition, the significant interaction between year and Athena SWAN accreditation/award suggested compared to universities of Bronze awardees, with time increased, the QS overall scores among HEIs with Silver awards has increased in lower rates (β=-0.77, p <.05). Considering the nonlinear effect (Appendix 3), results showed substantial decrease in QS overall scores in 2015/16 and 2016/17, compared to 2012/13. There was only one non-recipient Charter institution covered in QS rankings across the study period (i.e., London School of Economics and Political Science, top 37 institution globally), suggesting that most of the institutions in the UK considered to be globally elite have applied for and received Athena SWAN awards, with the highest-ranked institutions mostly having attained Silver awards.

[INSERT TABLE 4 HERE]

Discussion

A central goal of Athena SWAN is to move HEIs and the sector towards gender equity. The rising trends in gender diversity found in this study are generally encouraging. In all levels of Athena SWAN affiliation, and even in those without affiliation, gender diversity of managerial leaders and non-managerial professors

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increased over the five years examined here (2012/13-2016/17). In particular, we identified a faster rate of increase among managerial leaders, schools with Silver award, and non-recipient institutions compared to those not affiliated with Athena SWAN. During the same period of time, the average QS overall scores have increased from 2012/13 to 2015/16 but later decreased among both Bronze and Silver awardees. Institutions with Silver awards also had better university performance than those with Bronze awards. This is the first study that quantitatively examined the association between different levels of Athena SWAN awards/accreditation, gender diversity, and university performance.

For female managerial leaders, non-recipients showed greater female representation than Athena SWAN award recipients and non-Athena SWAN institutions. An examination of the 18 non-recipient institutions included in the current study showed that they joined the Athena SWAN Charter later than 2015/16,[38] which left them shorter time to meet the award criteria, even though they have high level of female representation initially. Athena awardees, importantly, showed higher growth rate of female managerial leader representation than non-Athena institutions. Such increasing rates even became faster among Athena institutions, suggesting the positive effect of involving in the Athena SWAN agenda on promoting gender equity. Notably, Silver awardees pushed past the point of one-third of managerial leaders being female, a figure previously suggested to be an essential milestone—namely 30%—for leadership board diversity affecting organizational performance.[39]

For non-managerial professors, the findings are mixed. Regardless of Athena SWAN accreditations/levels, less than one-third of such professors were women at the end of the 5-year period examined in this study (i.e., 2012/13-2016/17). Although all four groups moved towards greater gender diversity, the rates of growth were

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lower than those among female leaders, which is striking, given that around half of postgraduate research students are female.[40] The slow increase can be partly explained by the fact that while Athena SWAN awards may have impact on the recruitment of junior female faculty members, and it may be difficulty or take a longer period of time for junior faculty members to attain the title of professor. Other potential factors, such as the gender makeup of students graduating in different subject areas (e.g., liberal arts vs. STEM subjects), may also affect the speed of progress towards gender diversity for professors. Future studies should consider how gender differences in subject areas may influence progress towards gender diversity. Nevertheless, while non-Athena institutions had greater percentages of female non-managerial professors, HEIs with all levels of Athena SWAN accreditation/awards showed greater growth rates of female representation over time, suggesting the positive effects of applying and receiving the Athena awards on gender diversity among professors over time.

Having an Athena SWAN award, especially Silver award, is associated with a higher global ranking of UK HEIs measured by the QS ranking overall scores. As suggested in prior research,[22] this can be partly explained by institutions most affected by the NIHR funding being more motivated to pursue Silver awards. Higher-ranking institutions may also have better financial resources to commit to the Athena SWAN initiative. Such factors have contributed to debate about whether awards are recognizing what they ought to achieve.[1] Future studies examining the effect of Athena SWAN awards shall benefit from considering more dimensions of university performances and additional financial incentives that institutions may have for achieving gender equity.

Strengths and Limitations

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This study is the first quantitative study to examine gender diversity in managerial leadership and senior academic professor positions across UK HEIs with different Athena SWAN status in a longitudinal design. It adds to the existing research literature, which almost exclusively has a qualitative focus,[3 28] through quantitatively assessing the relationship between these variables of interest.

We are mindful of the following limitations of this study. First, our study is advantageous to obtain relatively large sample across five years; however, it is possible that the effect of Athena SWAN may differ for longer periods. As suggested by latest evaluation studies, Athena SWAN is a complex intervention and could influence changes in female representation in different ways and mechanisms.[23 29] Thus, it is possible that substantial increases in female representation, especially at senior academic levels, will require more time.[29] Future studies will do well to continue collecting data with longer periods with more data points and award details to examine the effect of Athena SWAN on gender diversity.

Second, besides quantitative analysis, qualitative studies examining how and why Athena SWAN might influence gender equality progress should be conducted to gain more understanding in its effect and offer useful suggestions for future practices.

Third, gender diversity in HEIs is affected by various social contextual factors (e.g., culture, GDP, policy, geography).[2-4 6 11 28 41] As the first quantitative study to explore the effect of Athena SWAN accreditation/awards, we did not control for these variables in the current analysis. Future research should consider these factors in analyzing the effect of Athena SWAN.

Fourth, prior to 2015, Athena SWAN focused primarily on academic and research staff,[25] while the HESA data used in this study covers five years from 2012/13 to 2016/17, and the contract levels A-E2 (defined as Leaders) include non-

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academic staff (e.g., Vice-Chancellor/Principal/equivalent; Director of major function/group of functions). Hence, the changes in the levels of gender diversity in the leadership position may be only indirectly attributable to Athena SWAN.

Fifth, the professor category in this study refers to senior academic rank (e.g., professors without line management responsibilities) and might have lower mobility than lower-rank faculty categories (e.g., assistant professors). It is possible that the impact of Athena SWAN is greater among lower academic ranks, which were outside the scope of the current study. Future studies can include data of female representation among lower-rank faculty positions and conduct a more comprehensive analysis of academics at different ranks.

Sixth, QS world ranking overall score was selected as a proxy indicator for university performance in this study and has some limitations related to using subjective ratings on institutional reputation, weightings, and possibly biased survey to build a composite indicator.[42] Future studies shall replicate the study using a suite of indicators with more stability and robustness.

Lastly, this study is based on HEIs with available employment data from HESA in the UK; the generalizability of its findings to UK HEIs and to other countries is yet to be determined. Future studies can also explore how other gender equity programs, such as Organizational Change for Gender Equity in STEM Academic Professions (ADVANCE) in the US, may affect gender diversity.

Conclusion

Under-representation of women in senior positions in higher education institutions has been widely identified as a major challenge to tackle, both in the UK and internationally. This study has revealed how Athena SWAN levels are related to gender diversity in higher education institutions. Despite overall positive trends

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towards gender diversity and positive associations between Athena SWAN level and increasing gender diversity for managerial leadership over time, the findings for full-time senior academics (i.e., professors without managerial duties) indicate that Athena SWAN may need to look closely at whether receiving the award actually brings a meaningful and sustained commitment to gender diversity as intended. The relationship between research funding and criteria for achieving Athena SWAN levels of accreditation/award warrants further attention to ensure appropriate institutional recognition and incentives. Finally, to address gender diversity in a broader cultural context and the degree to which the climate of the higher education environment is supportive of diversity, Athena SWAN could consider incorporating social marketing initiatives targeting attitudinal change, as shown in other domains, for instance with the Time to Change campaign.[43]

Figure 1: Changes of Female Representation in Leaders by Athena SWAN Levels from 2012/13 to 2016/17.

Figure 2: Changes of Female Representation in Professors by Athena SWAN Levels from 2012/13 to 2016/17.

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Table 1

Trends of female representation and QS overall scores from 2012/13 to 2016/17[a.]

Year	Female Representation												QS Overall Scores[c.]			
	Leaders (%)						Professors (%)						All HEIs	Bronze	Silver	Test Statistics
	N	All	Bronze	Silver	Non Recipient	Non-Athena	*N*	All	Bronze	Silver	Non Recipient	Non-Athena	*N*	Mean (*SD*)	Mean (*SD*)	Mean (*SD*)	
2012/13	136	37.2	35.5	28.7	43.5	41.2	100	24.2	24.3	18.9	26.6	31.4	47	57.8 (20.9)	53.0 (18.9)	70.1 (21.5)	*t* (45) = -2.67*, *d* = -.87
2013/14	136	37.2	35.0	29.9	44.3	41.5	106	24.6	24.4	19.5	26.4	32.2	46	60.2 (21.2)	55.4 (19.6)	72.5 (20.7)	*t* (44) = -2.63*, *d* = -.86
2014/15	137	38.0	36.8	30.3	44.5	40.5	103	25.3	24.9	20.8	28.0	36.7	47	61.3 (21.0)	56.6 (19.4)	73.6 (20.6)	*t* (45) = -2.65*, *d* = -.86
2015/16	135	38.7	37.2	33.4	46.7	40.1	102	26.1	25.3	21.9	32.9	33.9	46	63.2 (18.7)	59.6 (17.2)	72.3 (20.1)	*t* (44) = -2.16*, *d* = -.71
2016/17	133	39.5	37.9	34.1	47.4	41.2	103	26.8	26.4	22.3	31.9	32.1	46	56.9 (20.0)	52.9 (17.8)	67.3 (22.1)	*t* (45) = -2.26*, *d* = -.74
Rate Growth[b.]		8.4	9.0	20.3	11.1	2.4		7.6	5.7	14.3	17.5	1.1		1.6	3.1	-1.9	

Note. SD=Standard deviation, *t*-test was used to compare mean differences between Bronze and Silver awardees, *d*=Cohen's *d*.

[a.] Rates based on the availability of data in each year.

[b.] Methods of calculating the rate growth can be found in the Method section.

[c.] Analysis based on institutions with Bronze or Silver awards during the study periods.

* $p<0.05$.

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Table 2

Linear mixed effect models of female representation in leaders and professors from 2012/13 to 2016/17

	Among Athena Charter Members				Among All HEIs[d]			
	Leaders		Professors		Leaders		Professors	
	ß	95% CI	*ß*	95% CI	*ß*	95% CI	*ß*	95% CI
N	111		111		148		148	
Fixed effect								
Intercept[a]	0.32***	[0.25, 0.38]	0.20***	[0.15, 0.25]	0.28***	[0.22, 0.34]	0.22***	[0.15, 0.29]
Athena Award								
Non-Athena					(Ref)		(Ref)	
Non-Recipients	(Ref)		(Ref)		0.050*	[0.012, 0.095]	-0.005	[-0.077, 0.067]
Bronze	-0.062*	[-0.12, -0.002]	-0.040	[-0.11, 0.032]	-0.009	[-0.065, 0.047]	-0.048	[-0.134, 0.038]
Silver	-0.112**	[-0.189, -0.035]	-0.083	[-0.17, 0.007]	-0.057	[-0.132, 0.018]	-0.089	[-0.194, 0.016]
Year	0.009***	[0.006, 0.013]	0.006***	[0.004, 0.009]	0.008***	[0.005, 0.012]	0.006***	[0.003, 0.008]
Duration of Award[b]	-0.007	[-0.018, 0.005]	-0.002	[-0.016, 0.012]	-0.007	[-0.019, 0.005]	-0.002	[-0.017, 0.014]
Random effect[c]								
Intercept variance	0.059***		0.036***		0.076***		0.043***	
Year variance	-0.0037***		-0.002***		-0.005***		-0.002***	
Residual variance	0.001***		0.0004***		0.0011***		0.0004***	
AIC	-1615.6		-1869.4		-2048.3		-1924.3	
BIC	-1591.2		-1845.1		-2018.3		-1894.3	

Note. AIC = Akaike Information Criteria; BIC = Bayesian Information Criteria; 95% CI = 95% Confidence Interval.

[a.] Intercept coefficient estimates the mean initial percentages of female leaders or professors among the non-recipients (in the model among Athena Charter members) and non-Athena HEIs (in the model among all HEIs).

[b] Duration of being awarded Bronze or Silver during the study period (2012/13-2016/17).

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[c] Intercept variance indicates the between-group variances across HEIs; year invariance indicates the structured variance across years; residual variance indicates the variability of individual HEI's female representaion around the individual regression lines for each subjects.

[d.] All high education institute in the study, including non-Athena charter members, non recipient of Athena award, Bronze, and Silver awardees.

* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

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Table 3

Linear mixed effect models of female representation in leaders and professors from 2012/13 to 2016/17

	Among Athena Charter Members				Among All HEIs[d]			
	Leaders		Professors		Leaders		Professors	
	ß	95% CI	*ß*	95% CI	*ß*	95% CI	*ß*	95% CI
N	111		111		148		148	
Fixed effect								
Intercept[a]	0.45***	[0.42, 0.49]	0.30***	[0.26, 0.34]	0.40***	[0.38, 0.43]	0.31***	[0.25, 0.36]
Athena Award								
Non-Athena					(Ref)		(Ref)	
Non-Recipients	(Ref)		(Ref)		0.05*	[0.01, 0.093]	-0.01	[-0.08, -0.066]
Bronze	-0.06*	[-0.12, -0.002]	-0.04	[-0.11, 0.032]	-0.01	[-0.07, 0.046]	-0.05	[-0.13, 0.038]
Silver	-0.11**	[-0.19, -0.03]	-0.08	[-0.17, 0.0073]	-0.06	[-0.13, 0.02]	-0.09	[-0.20, 0.014]
Year (Centered)	0.011*	[0.0025, 0.020]	0.0070	[-0.0008, 0.015]	0.0039	[-0.0035, 0.0112]	-0.005	[-0.017, 0.0071]
Year*Athena Award								
Year*Non-Athena					(Ref)		(Ref)	
Year*Non-recipient	(Ref)		(Ref)		0.008	[-0.0045, 0.020]	0.012	[-0.003, 0.026]
Year*Bronze	-0.0034	[-0.013, 0.0065]	-0.0013	[-0.0096, 0.0069]	0.0042	[-0.0044, 0.013]	0.011	[-0.0018, 0.023]
Year*Silver	0.0016	[-0.012, 0.015]	0.0022	[-0.0083, 0.013]	0.0093	[-0.0042, 0.023]	0.014*	[0.0001, 0.023]
Duration of Award[b]	-0.0067	[-0.018, 0.0051]	-0.0018	[-0.016, 0.012]	-0.0067	[-0.019, 0.0053]	-0.0016	[-0.017, 0.014]
Random effect[c]								
Intercept variance	0.005***		0.006***		0.005***		0.008***	
Year variance	0.0001		0.0001		-0.0001		-0.00003	
Residual variance	0.001***		0.0004***		0.001***		0.0004***	

AIC	-1612.6	-1866.3	-2044.8	-1922.2
BIC	-1582.8	-1836.5	-2005.8	-1883.2

Note. AIC = Akaike Information Criteria; BIC = Bayesian Information Criteria; 95% CI = 95% Confidence Interval.

[a.] Intercept coefficient estimates the mean initial percentages of female leaders or professors among the non-recipients (in the model among Athena Charter members) and non-Athena HEIs (in the model among all HEIs).

[b.] Duration of being awarded Bronze or Silver during the study period (2012/13-2016/17).

[c] Intercept variance indicates the between-group variances across HEIs; year invariance indicates the structured variance across years; residual variance indicates the variability of individual HEI's female representaion around the individual regression lines for each subjects.

[d.] All high education institute in the study, including non-Athena charter members, non recipient of Athena award, Bronze, and Silver awardees.

* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

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Table 4

Linear mixed effect models of QS scores in leaders and professors from 2012/13 to 2016/17

	QS Overall Scores[d]			
	Model 1		Model 2	
	ß	95% CI	*ß*	95% CI
N	93		93	
Fixed effect				
Intercept[a]	63.55***	[32.19, 94.91]	61.03***	[30.34, 91.72]
Athena Award (Ref. Bronze)				
Silver	11.80*	[0.01, 23.59]	16.13**	[3.73, 28.53]
Year (Centered)	-0.085	[-0.40, 0.23]	0.13	[-0.24, 0.50]
Year*Athena Award (Ref. Year*Bronze)				
Year*Silver			-0.77*	[-1.45, -0.08]
Duration of Award[b]	-1.24	[-7.53, 5.06]	-1.23	[-7.52, 5.06]
Random effect[c]				
Intercept variance	559.04***		361.96***	
Year variance	-6.537		-6.445	
Residual variance	11.504***		11.21***	
AIC	1471.6		1468.8	
BIC	1489.3		1489.0	

Note. AIC = Akaike Information Criteria; BIC = Bayesian Information Criteria; 95% CI = 95% Confidence Interval.

[a.] Intercept coefficient estimates of the mean initial QS score among Bronze Awardees.

[b.] Duration of being awarded Bronze or Silver during the study period (2012/13-2016/17).

[c] Intercept variance indicates the between-group variances across HEIs; year invariance indicates the structured variance across years; residual variance indicates the variability of individual HEI's female representaion around the individual regression lines for each subjects.

[d.] Only conducted among Bronze, and Silver awardees.

* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

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Contribution

Conception and design of study: YX, EP, TKFA, PSFY. Data collection: EP, YX. Data analysis: YX. Interpretation of data: YX, EP, PSFY, TKFA. Drafting the manuscript: YX, EP. Revising the manuscript critically for important intellectual content: YX, EP, TKFA, PSFY. Approval of the version of the manuscript to be published: YX, EP, TKFA, PSFY.

Competing interests: None declared.

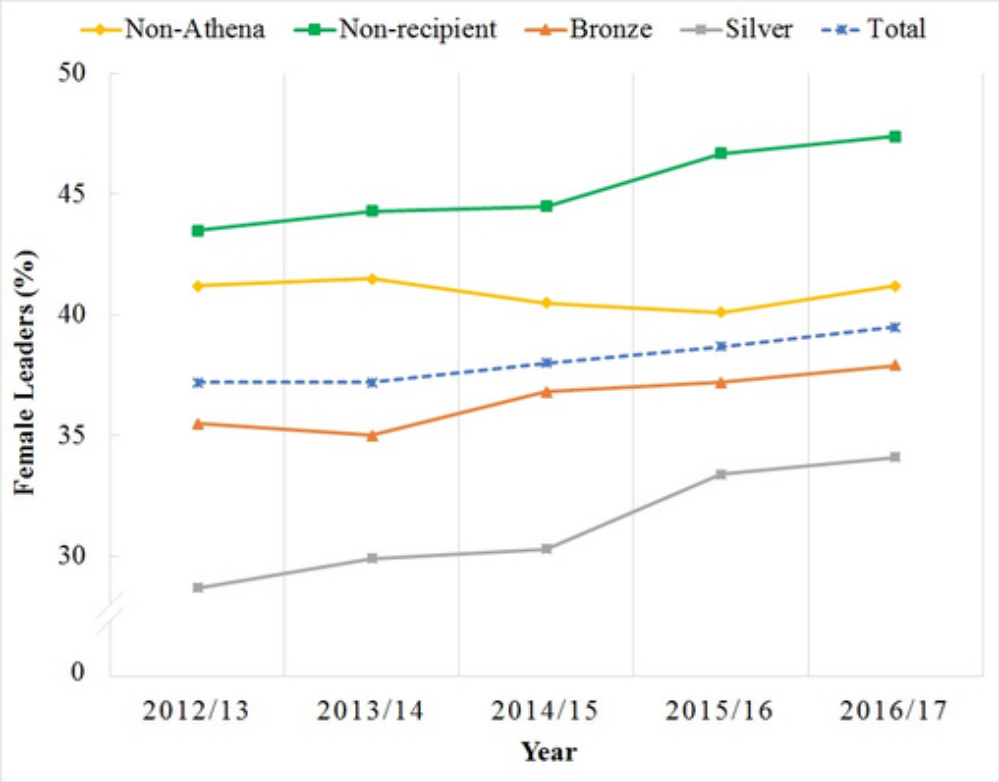
Funding: This research received funding TKFA's Karen Lo Eugene Chuang Professorship in Diversity and Equity at the University of Hong Kong.

Data availability statement: Data are available upon reasonable request and when the dissemination of the project full results is completed.

Patient consent for publication: Not required.

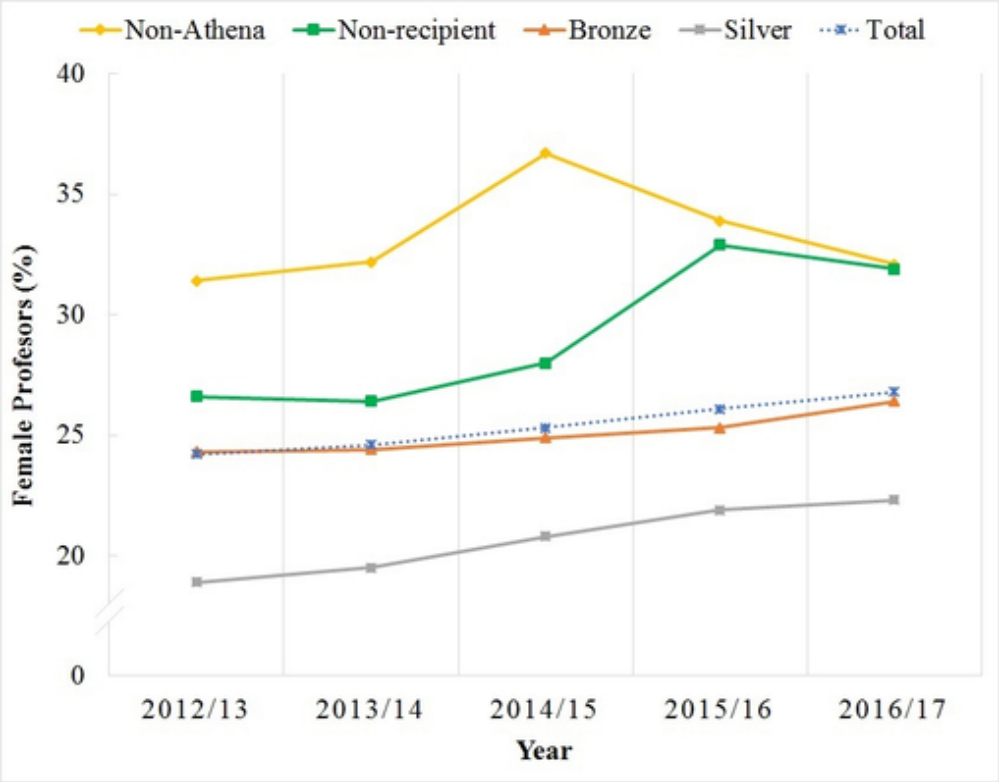
Provenance and peer review: Not commissioned; externally peer reviewed.

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Changes of Female Representation in Leaders by Athena SWAN Levels from 2012/13 to 2016/17.

21x16mm (600 x 600 DPI)



Changes of Female Representation in Professors by Athena SWAN Levels from 2012/13 to 2016/17.

21x16mm (600 x 600 DPI)

Appendix 1

Linear mixed effect models of female representation in leaders and professors from 2012/13 to 2016/17

	Among Athena Charter Members				Among All HEIs[d]			
	Leaders		**Professors**		**Leaders**		**Professors**	
	ß	95% CI	ß	95% CI	ß	95% CI	ß	95% CI
N	111		111		148		148	
Fixed effect								
Intercept[a]	0.44***	[0.41, 0.47]	0.29***	[0.25, 0.33]	0.39***	[0.37, 0.41]	0.30***	[0.24, 0.35]
Athena Award								
Non-Athena					(Ref)		(Ref)	
Non-Recipients	(Ref)		(Ref)		0.052*	[0.011, 0.093]	-0.0069	[-0.079, 0.065]
Bronze	-0.062*	[-0.12, -0.0019]	-0.041	[-0.11, 0.031]	-0.0093	[-0.066, 0.047]	-0.048	[-0.13, 0.040]
Silver	-0.110**	[-0.19, -0.034]	-0.084	[-0.17, 0.0062]	-0.059	[-0.13, 0.016]	-0.090	[-0.20, 0.015]
Year								
2012/13	(Ref)		(Ref)		(Ref)		(Ref)	
2013/14	0.00038	[-0.011, 0.011]	0.001	[-0.0062, 0.0085]	0.001	[-0.0092, 0.011]	0.002	[-0.0056, 0.0091]
2014/15	0.014*	[0.0031, 0.025]	0.0084*	[0.0011, 0.016]	0.012*	[0.0017, 0.022]	0.0079*	[0.00054, 0.015]
2015/16	0.026***	[0.015, 0.037]	0.016***	[0.0084, 0.023]	0.022***	[0.011, 0.032]	0.014***	[0.0067, 0.022]
2016/17	0.034***	[0.023, 0.045]	0.023***	[0.016, 0.030]	0.030***	[0.020, 0.040]	0.022***	[0.014, 0.029]
Duration of Award[b]	-0.0068	[-0.019, 0.0051]	-0.0017	[-0.015, 0.012]	-0.0068	[-0.019, 0.0052]	-0.0017	[-0.017, 0.014]
Random effect[c]								
Intercept variance	0.005***		0.006***		0.005***		0.008***	
Residual variance	0.002***		0.001***		0.002***		0.001***	
AIC	-1565.2		-1772.9		-1959.8		-1817.7	

BIC	-1538.1	-1745.8	-1926.9	-1784.7

Note. AIC = Akaike Information Criteria; BIC = Bayesian Information Criteria; 95% CI = 95% Confidence Interval.

[a.] Intercept coefficient estimates the mean initial percentages of female leaders or professors among the non-recipients (in the model among Athena Charter members) and non-Athena HEIs (in the model among all HEIs).

[b.] Duration of being awarded Bronze or Silver during the study period (2012/13-2016/17).

[c] Intercept variance indicates the between-group variances across HEIs; year invariance indicates the structured variance across years; residual variance indicates the variability of individual HEI's female representaion around the individual regression lines for each subjects.

[d.] All high education institute in the study, including non-Athena charter members, non recipient of Athena award, Bronze, and Silver awardees.

* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

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Appendix 2

Linear mixed effect models of female representation in leaders and professors from 2012/13 to 2016/17

	Among Athena Charter Members				Among All HEIs[d]			
	Leaders		Professors		Leaders		Professors	
	ß	95% CI	ß	95% CI	ß	95% CI	ß	95% CI
N	111		111		148		148	
Fixed effect								
Intercept[a]	0.44***	[0.40, 0.47]	0.29***	[0.25, 0.33]	0.40***	[0.37, 0.42]	0.31***	[0.25, 0.37]
Athena Award								
Non-Athena					(Ref)		(Ref)	
Non-Recipients	(Ref)		(Ref)		0.038	[-0.01, 0.08]	-0.023	[-0.10, 0.06]
Bronze	-0.05	[-0.12, 0.01]	-0.04	[-0.11, 0.034]	-0.016	[-0.07, 0.04]	-0.063	[-0.15, 0.03]
Silver	-0.11**	[-0.20, -0.03]	-0.09	[-0.18, 0.0002]	-0.076	[-0.15, 0.002]	-0.114*	[-0.22, -0.01]
Year								
2012/13	(Ref)		(Ref)		(Ref)		(Ref)	
2013/14	0.01	[-0.02, 0.04]	0.002	[-0.02, 0.02]	0.003	[-0.02, 0.03]	0.007	[-0.024, 0.038]
2014/15	0.01	[-0.02, 0.04]	0.009	[-0.01, 0.03]	0.004	[-0.02, 0.03]	-0.006	[-0.040, 0.028]
2015/16	0.04**	[0.01, 0.06]	0.020	[-0.003, 0.04]	0.007	[-0.02, 0.03]	-0.028	[-0.064, 0.008]
2016/17	0.04**	[0.02, 0.07]	0.018	[-0.005, 0.040]	0.016	[-0.01, 0.04]	-0.024	[-0.063, 0.014]
Year*Athena Award					(Ref)		(Ref)	
2012/13* Non-Athena					(Ref)		(Ref)	
2013/14* Non-Athena					(Ref)		(Ref)	

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2014/15* Non-Athena					(Ref)		(Ref)	
2015/16* Non-Athena					(Ref)		(Ref)	
2016/17* Non-Athena					(Ref)		(Ref)	
2012/13* Non-Recipients	(Ref)		(Ref)		(Ref)		(Ref)	
2013/14* Non-Recipients	(Ref)		(Ref)		0.006	[-0.03, 0.04]	-0.006	[-0.044, 0.033]
2014/15* Non-Recipients	(Ref)		(Ref)		0.007	[-0.03, 0.04]	0.015	[-0.025, 0.056]
2015/16* Non-Recipients	(Ref)		(Ref)		0.031	[-0.005, 0.07]	0.047*	[0.005, 0.090]
2016/17* Non-Recipients	(Ref)		(Ref)		0.028	[-0.008, 0.06]	0.041	[-0.003, 0.086]
2012/13* Bronze	(Ref)		(Ref)		(Ref)		(Ref)	
2013/14* Bronze	-0.013	[-0.04, 0.02]	-0.001	[-0.02, 0.02]	-0.007	[-0.03, 0.02]	-0.007	[-0.039, 0.026]
2014/15* Bronze	0.004	[-0.03, 0.03]	-0.003	[-0.03, 0.02]	0.011	[-0.015, 0.04]	0.012	[-0.023. 0.047]
2015/16* Bronze	-0.02	[-0.05, 0.01]	-0.007	[-0.03, 0.02]	0.014	[-0.012, 0.04]	0.041*	[0.004, 0.077]
2016/17* Bronze	-0.02	[-0.05, 0.01]	0.004	[-0.02, 0.03]	0.012	[-0.014, 0.038]	0.046*	[0.007, 0.085]
2012/13* Silver	(Ref)		(Ref)		(Ref)		(Ref)	
2013/14* Silver	0.003	[-0.04, 0.04]	0.004	[-0.03, 0.03]	0.008	[-0.03, 0.05]	-0.002	[-0.039, 0.036]
2014/15* Silver	0.001	[-0.04, 0.04]	0.010	[-0.02, 0.04]	0.007	[-0.03, 0.05]	0.025	[-0.014, 0.064]
2015/16* Silver	0.001	[-0.04, 0.05]	0.010	[-0.02, 0.04]	0.036	[-0.004, 0.08]	0.058**	[0.017, 0.099]
2016/17* Silver	0.005	[-0.04, 0.05]	0.016	[-0.01, 0.05]	0.033	[-0.007, 0.07]	0.058**	[0.014, 0.101]
Duration of Award[b]	-0.007	[-0.02, 0.01]	-0.002	[-0.02, 0.01]	-0.007	[-0.02, 0.01]	-0.002	[-0.017, 0.014]
Random effect[c]								
Intercept variance	0.005***		0.006***		0.005***		0.008***	
Residual variance	0.002***		0.001***		0.002***		0.001***	
AIC	-1555.4		-1760.7		-1946.2		-1809.5	
BIC	-1506.6		-1711.9		-1877.3		-1740.6	

Note. AIC = Akaike Information Criteria; BIC = Bayesian Information Criteria; 95% CI = 95% Confidence Interval.

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[a.] Intercept coefficient estimates the mean initial percentages of female leaders or professors among the non-recipients (in the model among Athena Charter members) and non-Athena HEIs (in the model among all HEIs).

[b.] Duration of being awarded Bronze or Silver during the study period (2012/13-2016/17).

[c] Intercept variance indicates the between-group variances across HEIs; year invariance indicates the structured variance across years; residual variance indicates the variability of individual HEI's female representaion around the individual regression lines for each subjects.

[d.] All high education institute in the study, including non-Athena charter members, non recipient of Athena award, Bronze, and Silver awardees.

* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

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Appendix 3

Linear mixed effect models of QS scores in leaders and professors from 2012/13 to 2016/17.

	QS Overall Scores[d]			
	Model 1		Model 2	
	ß	95% CI	ß	95% CI
N	93		93	
Fixed-effect				
Intercept[a]	58.00***	[26.11, 89.89]	57.64***	[25.77, 89.51]
Athena award levels				
Bronze	(Ref)		(Ref)	
Silver	16.09**	[3.86, 28.32]	17.34**	[5.06, 29.62]
Year				
2012/13	(Ref)		(Ref)	
2013/14	2.75***	[1.84, 3.65]	2.88***	[1.87, 3.89]
2014/15	3.52***	[2.62, 4.42]	3.53***	[2.53, 4.54]
2015/16	4.72***	[3.82, 5.63]	5.73***	[4.72, 6.74]
2016/17	-1.51**	[-2.41, -0.60]	-0.89	[-1.90, 0.13]
Year*Athena award				
2012/13* Bronze			(Ref)	
2013/14* Bronze			(Ref)	
2014/15* Bronze			(Ref)	
2015/16* Bronze			(Ref)	
2016/17* Bronze			(Ref)	
2012/13* Silver			(Ref)	
2013/14* Silver			-0.49	[-2.41, 1.42]
2014/15* Silver			-0.05	[-1.96, 1.86]
2015/16* Silver			-3.57***	[-5.48, -1.65]
2016/17* Silver			-2.19*	[-4.10, -0.27]
Duration of award[b]	-0.99	[-7.52, 5.54]	-0.99	[-7.52, 5.54]
Random effect[c]				
Intercept variance	353.90***		353.63***	
Residual variance	4.89***		4.40***	
AIC	1321.0		1309.3	
BIC	1343.8		1342.2	

Note. AIC = Akaike Information Criteria; BIC = Bayesian Information Criteria; 95% CI = 95% Confidence Interval.

[a.] Intercept coefficient estimates of the mean initial QS score among Bronze awardees.

[b.] All high education institute in the study, including non-Athena charter members, non recipient of Athena award, Bronze, and Silver awardees.

[c] Intercept variance indicates the between-group variances across HEIs; year invariance indicates the structured variance across years; residual variance indicates the variability of

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individual HEI's female representaion around the individual regression lines for each subjects.

[d.] Duration of being awarded Bronze or Silver during the study period (2012/13-2016/17).

* $p<0.05$, ** $p<0.01$, *** $p<0.001$.

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STROBE 2007 (v4) Statement—Checklist of items that should be included in reports of *cohort studies*

Section/Topic	Item #	Recommendation	Reported on page #
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	1-2
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	2-3
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	5-10
Objectives	3	State specific objectives, including any prespecified hypotheses	9-10
Methods			
Study design	4	Present key elements of study design early in the paper	2-3, 10-13
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	10-13
Participants	6	(*a*) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up	10-13
		(*b*) For matched studies, give matching criteria and number of exposed and unexposed	10-13
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	10-13
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	10-13
Bias	9	Describe any efforts to address potential sources of bias	10-13, 21-22
Study size	10	Explain how the study size was arrived at	10-13
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	10-13
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	13-15
		(*b*) Describe any methods used to examine subgroups and interactions	13-15
		(*c*) Explain how missing data were addressed	13-15
		(*d*) If applicable, explain how loss to follow-up was addressed	/
		(*e*) Describe any sensitivity analyses	/
Results			

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Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	15-18
		(b) Give reasons for non-participation at each stage	/
		(c) Consider use of a flow diagram	/
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	15
		(b) Indicate number of participants with missing data for each variable of interest	29
		(c) Summarise follow-up time (eg, average and total amount)	
Outcome data	15*	Report numbers of outcome events or summary measures over time	15-18
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	15-18
		(*b*) Report category boundaries when continuous variables were categorized	/
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	/
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	15-18
Discussion			
Key results	18	Summarise key results with reference to study objectives	18-19
Limitations			
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	20-22
Generalisability	21	Discuss the generalisability (external validity) of the study results	22
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	36

*Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.

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